An 8T SRAM Based Digital Compute-In-Memory Macro For Multiply-And-Accumulate Accelerating

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***Abstract*—Compute-in-memory (CIM) has been a promising technology to reduce the data movement energy and latency, which is the bottleneck of Von Neumann architecture. Digital approaches in CIM macro have many advantages compared with analog counterparts, such as programmability and inference precision. However, previous works with digital approaches generally employ complex SRAM bit-cells and computational components, which cause a large area overhead. In this paper, we propose a new 8T SRAM bit-cell to reduce the overall area of the SRAM array, which is able to implement the 1-bit multiplication without read-disturb issue. Additionally, the interleaving adder tree and dual supply voltage strategy are employed for further reduction on area and power consumption of computational circuits. Besides, a result combination circuit is designed to increase the bit-precision flexibility. A 16Kb SRAM CIM macro with proposed techniques is designed in a 40-nm CMOS technology. The simulation results show that our work achieves 820 GOPS throughput and 94 TOPS/W energy efficiency with 4-b of both input and weight. It achieves 1.3× higher energy efficiency and 70% area reduction when compared to the recent state-of-the-art works.**

Keywords—Compute-in-memory, SRAM, Programmability, Digital approach.

I. INTRODUCTION

Artificial Intelligence (AI) and Internet of Things (IoT) applications have developed rapidly recent years. Devices are required to improve throughput and energy efficiency to meet the needs of these applications. Deep neural network (DNN) is one of the most successful algorithms in AI applications, so hardware accelerators that are applied to DNN have attracted the attention of many researchers. For AI edge device chip design, ultra-low power consumption and limited area are two critical challenges. In traditional Von Neumann architecture, computational unit and memory unit are separated. However, in today's advanced process technology, power consumption and latency in computation are much less than those in data movement between computational unit and memory [1], which is referred to as the issue of "Memory Wall". Thus, CIM has been proposed to solve this problem by enabling computing within the memory array, which makes it a promising approach to power reduction.

Typical DNN algorithms include huge amounts of multiply-and-accumulate (MAC) operations. The energy efficiency and performance of a system largely depend on the implementation of MAC operations. CIM architectures based on the SRAM are suitable for MAC operation because of the high parallelism of SRAM array. Recent SRAM-based CIM works mainly include analog and digital approaches. For analog approaches [2]-[4], the weights are generally stored in bit-cells. The multiplication result is generally represented by charge or discharge current caused by bit-cell's store node and the total current of bit lines (BLs) indicates the accumulated results of products. As the final results are represented by an analog signal, data conversion circuits such as Analog-Digital Converter (ADC) are indispensable for the subsequent signal processing. Many state-of-the-art CIM works choose analog approaches and these works are able to achieve very fast speed and high energy efficiency which are benefit from the character of analog computing. However, inference precision of DNN would decrease heavily due to the PVT (process, voltage, and temperature) variation and nonlinearity. Additionally, the overhead of data conversion circuits and low bit-precision are two critical challenges for analog approaches, and thus limit their use. Furthermore, 6T SRAM bit-cell which is widely used in analog approaches would incur the read-disturb issue because all rows need to be turned on simultaneously to achieve parallelism. Although it can be solved by using extra transistors acting as read buffers to isolate the store nodes and BLs, the area overhead will increase. Digital approaches [5]-[7] generally employ dedicated SRAM bit-cells such as 6T with an extra CMOS NOR gate to implement the multiplication. Besides, embedded computational logics are generally used in the SRAM array for parallel addition. As those SRAM bit-cells are able to output multiplication results represented by digital signals, the computational process is all-digital. Therefore, there is no inference precision loss as they are insensitive to noise and process variation. Moreover, digital works can be programmable and support alterable bit-precision to improve flexibility. While these works generally use complex bit-cells and computational circuits, which increase area overhead dramatically.

To reduce the area overhead, in this work we have proposed an all-digital CIM macro with a new pass transistor based 8T (PT-8T) SRAM bit-cell. An interleaving adder tree (IAT) is designed with two kinds of full adders (FAs) and a dual supply voltage strategy is used to achieve a better trade-off between area and performance. Also, the bit width of weight and input can be programmed so as to improve the flexibility. With the above design techniques, the proposed CIM macro is capable for high-efficiency MAC operations.

The remainder of this paper is organized as follows. Section II describes the proposed CIM macro architecture and detailed circuit design. Section III shows the simulation results and analysis. Section IV concludes our work.

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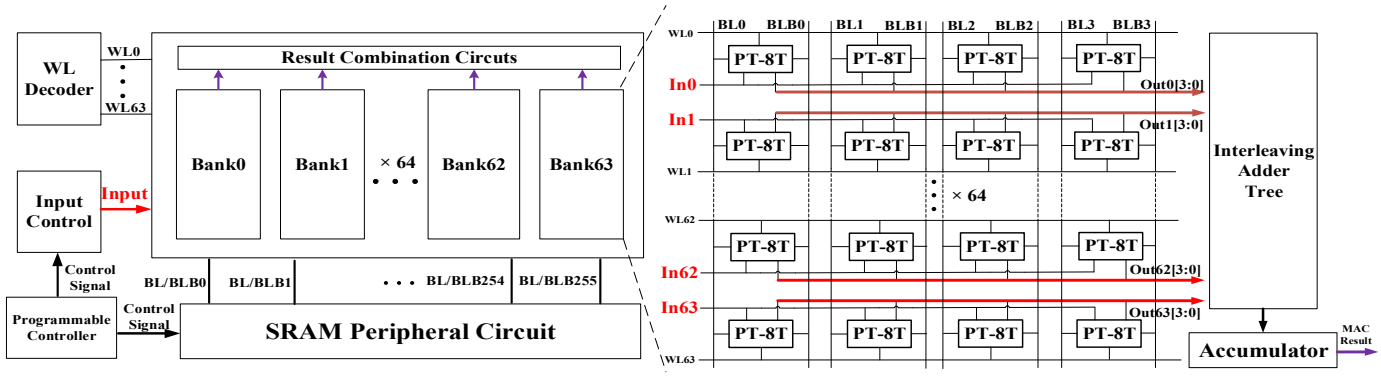


Fig. 1. Overall achitecture of the proposed CIM macro

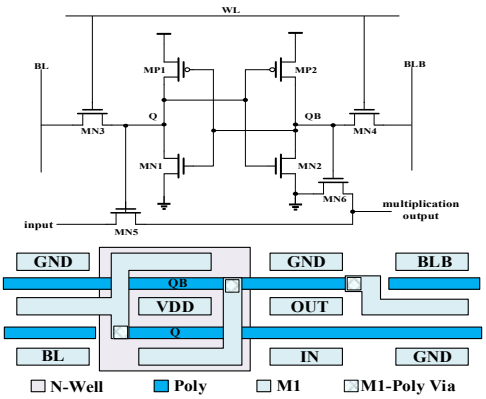


Fig. 2. The schematic and layout of pass transistor based 8T SRAM bit-cell.

II. CIM Architecture And Circuit Design

A. Overall Achitecture

Fig. 1 shows the overall architecture of the proposed CIM macro. The size of the SRAM array is 16Kb (64×256) and the array is divided into 64 banks. Each bank includes a 64×4 SRAM array, an IAT, and an accumulator. By configuring the programmable controller, the proposed CIM macro is able to operate in either SRAM mode or CIM mode.

In SRAM mode, the procedure of read and write operation is exactly the same as 6T SRAM. SRAM peripheral circuits contain typical components such as pre-charge circuits and Sense-Amplifier (SA). In CIM mode, the bit-width of input is arbitrary from 1b to 16b, and the weight can be either 4b or 8b. Input data which is controlled by the input controller would be fed into PT-8T bit-cells in a serial manner to generate multiplication results. In this step, there are 64 partial products with the same weight of multiplier (same bit of input) generated in one clock cycle. Then, IAT would compute the sum of them and deliver it to the accumulator. Finally, the accumulator computes the MAC result by shifting and adding the outputs of IAT in consecutive clock cycles.

B. Pass Transistor Based 8T SRAM Bit-Cell

Fig. 2 shows the schematic and layout of PT-8T SRAM bit-cell. It is similar to typical 6T SRAM except for the pass transistor logic (AND) that consists of MN5 and MN6. In order to reduce the effects of PVT variation, the size of MN5 is larger than MN6.

In SRAM mode, for read operation, BL and BLB are pre-charged before WL is asserted. The store node will discharge bit-line according to stored data and SA will capture the voltage drop in bit-line to output the result. For write operation, data to be written in is loaded in bit-lines. Upon the assertion of WL, bit-lines will conditionally discharge the store node.

In CIM mode, the weight stored in bit-cells will be multiplied by the input via the pass transistor logic. Different from many CIM works, WL does not need to be turned on, i.e., MN3 and MN4 are turned off during multiplication operation. Thus, there is no read-disturb issue in this design. Besides, compared with recent state-of-the-art digital works [5]-[7], PT-8T bit-cell is a simple and symmetric structure and merely uses 8 transistors. Comparing with the design in [7], which employs a CMOS NOR gate to implement multiplication, the area of the SRAM array in this work can be reduced significantly.

C. Interleaving Adder Tree

There are two kinds of FAs (28T and 10T) used in IAT as shown in Fig. 3. Between them, 28T FA is faster than 10T FA due to the CMOS logic and more stable as it is insensitive to noise. However, the area and power consumption of 28T FA is much larger than those of 10T FA.

To achieve the balance of power, performance and area, the IAT structure is proposed as shown in Fig.4. The two kinds of FA are arranged interleaved to ensure that the area and speed are both acceptable. Note that there is threshold voltage loss in 10T FA and the loss will be delivered to the next stage, which may cause malfunction. The interleaving structure could intercept the delivery of voltage loss as the 28T FA can restore back the voltage to supply rails. Notably, as the first stage (4-bit FA) in IAT requires the most amount of full adder cells, 128 (4×32) in total. So 10T FAs are arranged in the first stage for the purpose of area saving. While last several stages can use 28T FA to improve the performance with less area overhead as there are fewer 1-bit FAs.

D. Result Combination Circuit

In order to improve the flexibility of the bit precision of weight, we design the result combination circuit (RCC), which is consist of 32 pairs of adder and multiplexer, as shown in Fig.5. For 8-b weight, the higher 4 bits and lower 4 bits are stored in the same row of two adjacent banks. The MAC results of the two banks will be shifted and

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summed to produce the final MAC results. While for 4b weight, the shift and add logic should be bypassed.

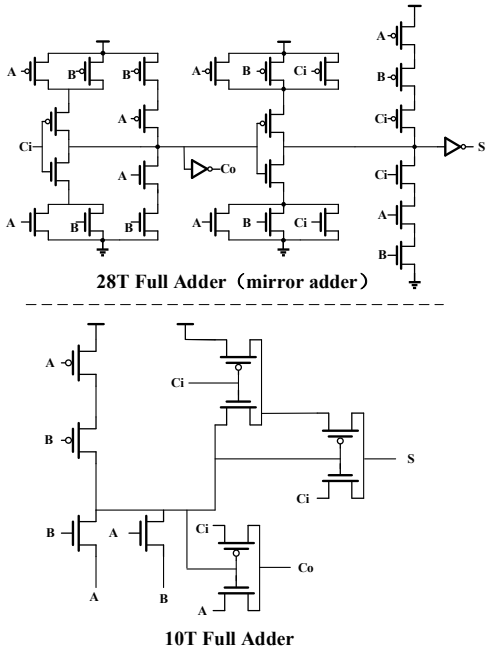


Fig. 3. The schematic of 28T full adder and 10T full adder.

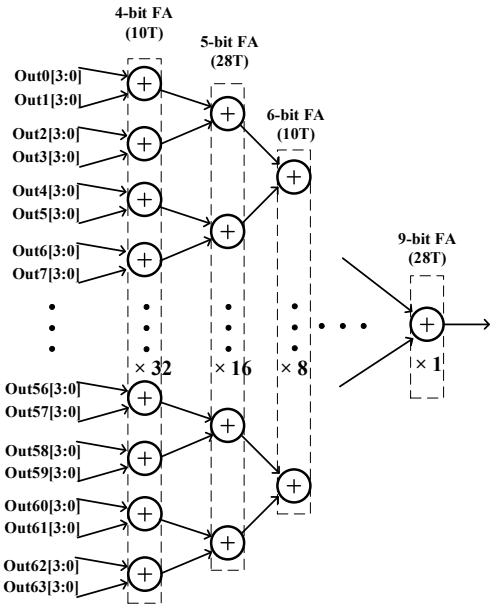


Fig. 4. The structure of interleaving adder tree (IAT).

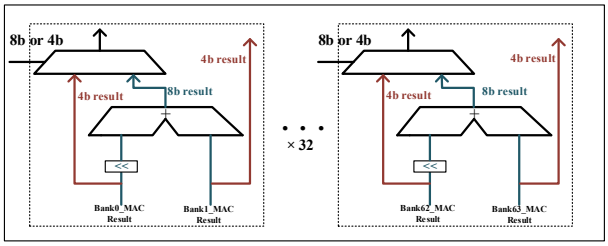


Fig. 5. The structure of result combination circuit (RCC)

E. Dual Supply Voltage Strategy

10T FA is able to reduce the area overhead with the sacrifice in stability and speed. The decrease of stability and speed will be severer when 10T FA works at a lower voltage. Thus, relatively high supply voltage is used for 10T FAs for better performance. While for the other part, their stability and speed are fine even with a lower supply voltage. Therefore, a dual supply voltage strategy is adopted in this design to reduce the power consumption and the delay time.

In the proposed CIM macro, there are two supply voltages, the higher one is for 10T FAs and read/write operations of SRAM to improve the performance, and the lower one is for the other circuits and hold operation of SRAM to reduce the power consumption. This dual supply voltage strategy follows the typical SRAM dual-rail design and adapts it to the computational part, so there is little additional overhead. The specific values of the voltages will be discussed in Section III.

F. Compute-In-Memory Operation

The bit-width of input can be arbitrary when the macro works in CIM mode. While the width of weight is either 4-b or 8-b. As Fig. 6(a) shows, input will be fed into the SRAM array every clock cycle in a serial manner, from MSB to LSB. The advantage of the serial manner is that we can easily control the bit width of input, so the application scope could be larger than many analog counterparts whose bit-width of input and weight are fixed.

The weights stored in bit-cells multiply (“AND”) with one bit of inputs by the pass transistor logic in PT-8T bit-cells and there are 64 partial products (4-bit) with the same weight generated in this step. These partial products will be added in IAT and the sum (10-bit) will be delivered to the accumulator. The structure of accumulator is shown in Fig. 6(b). The sum from the last clock cycle will be left shifted one bit and then add with the sum from the current clock cycle in the accumulator.

III. Experimental Results And Analysis

We have designed the proposed CIM macro in a 40-nm CMOS technology. The size of SRAM is 16Kb (64×256). In this section, we will show our simulation results and make comparison with several state-of-the-art works.

A. Delay of Interleaving Adder Tree

As the performance is largely determined by IAT delay, we get the average delay of IAT by simulating with random excitation. Fig. 7(a) shows the simulation results of IAT delay. The delay of IAT increases when the supply voltage drops.

Noticed that when 10T FA are used in 8-bit adders, the delay deteriorates severely compared to 28T FAs due to degraded driving capability caused by the pass transistor logic in 10T FA. At 1.0V/0.8V supply voltages, the delay has reached 2.33ns. Fortunately, considering that there are only two 8-bit adders in IAT (16 1-bit FAs in total) so the area overhead of this stage takes a small part of the entire IAT stage, 28T FAs are employed in 8-bit adders to achieve a great increase of performance with a small area cost. The delay of IAT is 5.58ns at 1.0V/0.8V supply voltage.

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TABLE I. COMPARISON RESULT WITH RELATED WORKS

	ISCAS' 21[2]	TCAS-II' 21[3]	TCAS-I' 19[4]	ISSCC' 19[5]	ESSCIRC' 19[6]	ISSCC' 21[7]	***This Work***
Technology	65nm	55nm	65nm	28nm	65nm	22nm	**40nm**
MAC approach	Analog	Analog	Analog	Digital	Digital	Digital	**Digital**
Array Size	16Kb	216Kb	4Kb	128Kb	16Kb	64Kb	**16Kb**
Supply Voltage	1.0V	0.4V/0.2V	0.8V-1.2V	0.6-1.1V	0.6-0.8V	0.72V	**1.0V/0.8V**
Cell Type	AND8T	13T	DSC6T	8T	6T	6T	**PT-8T**
Cell Area(Normalized)	3.3	1.5	0.9	1.1	2.7	1.3	**1**
Input Bit Precision	4b	1b	1b	Arbitrary	1-16b	1-8b	**1-8b**
Weight Bits Precision	4b	1b	1b	Arbitrary	4/8/12/16b	4/8/12/16b	**4/8b**
Frequency	50MHz	98MHz	435MHz	475MHz	138MHz	100MHz	**100MHz**
Throughput(GOPS)	/	913	278.2	9.2	567 (1b/1b)	3300 (4b/4b)	**820 (4b/4b)**
Energy Efficiency(TOPS/W)	69.33	5526	30.49-55.8	0.55	117.3 (1b/1b)	89 (4b/4b) 24.7(8b/8b)	**94 (4b/4b)** **23.8(8b/8b)**

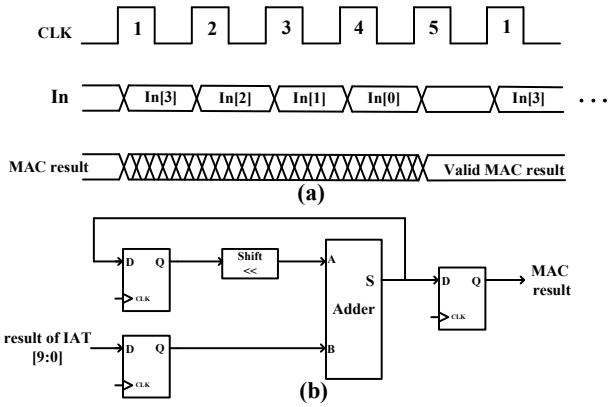


Fig. 6. (a) Timing diagram. (b) The structure of accumulator.

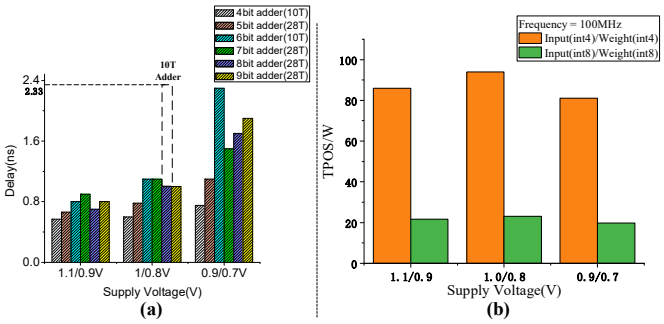


Fig. 7. (a) The simulation results of interleaving adder tree (IAT) delay. (b) The simulation results of energy effciency of proposed Compute-in-memory macro.

B. Energy Efficiency

Fig. 7(b) shows the energy efficiency simulation results of our work at different supply voltages. At 100MHz frequency, when the bit width of input and weight are both 4-bit, the highest energy efficiency is 94 TOPS/W at 1.0V/0.8V supply voltage. At 0.9V/0.7V supply voltages, the energy efficiency decreases due to the increased delay and leakage power in the SRAM array. While at 1.1V/0.9V supply voltages, the throughput get little improvement, which makes the energy efficiency also lower than that at 1.0V/0.8V.

C. Comparison

Table I shows the comparison results of our work and several state-of-the-art works. We have carried out a multi-faceted analysis, including technology, MAC approach (digital or analog), array size, supply voltage, cell type and area, widths of input and weight, frequency, throughput, and energy efficiency. Compared to works with digital approaches [5]-[7], due to the reduced area of PT-8T bit-cell in SRAM array and 10T FA in IAT, our work exhibits the smallest area overhead while with a small loss in throughput and energy efficiency. Moreover, compared to works with analog approaches [2]-[4], our design achieves relatively high energy efficiency because there is no overhead from data conversions between analog and digital domains. Moreover, there is no inference precision loss in this work because process variation and noise which are critical and knotty in analog approaches are inessential in our approaches.

IV. CONCLUSION

We propose several effective techniques to improve energy efficiency and reduce the area of CIM macro with digital approaches. First, we propose a PT-8T bit-cell that is able to perform the multiplication operation within the bit-cell without the read-disturb and pseudo-write issue. Due to the simple and symmetric structure, PT-8T bit-cell costs less area overhead compared to most bit-cells used in previous works. Second, we propose an interleaving adder tree and employ the dual supply voltage strategy to achieve a better trade-off. Finally, we designed the RCC to improve the bit precision flexibility of weight for wider application range. The simulation results show that our design, with less area overhead, has the promising throughput and energy efficiency of 820 GOPS and 94 TOPS/W (4-bit input and 4-bit weight), respectively.

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